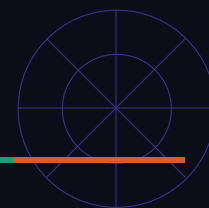


GAME DESIGN DOCUMENT

THE WEAVER

A dimensional platformer where a lone Weaver crosses collapsing dimensions to find the source of the Great Tear — and choose the fate of the multiverse.

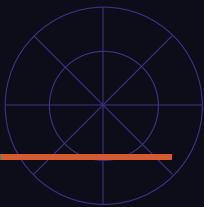
Team	GameCloud
Engine	Godot 4
Platform	Mobile — iOS / Android (Portrait)
Genre	2D Platformer · Environmental Puzzle · Exploration
Art Style	Illustration / Cartoon — African-Inspired Palette
Jam Theme	Play Without Borders
Prototype Scope	1 complete dimension — L'Inversé
Run Duration	15–20 min first run · 4–5 min speedrun



◆ TEAM — GAMECLOUD

Role	Pole	Responsabilites Jam
Game Designer 1	Interface et UX	Wireframes UI · MainMenu.tscn · PauseMenu.tscn · Settings · polish boutons et transitions
Game Designer 2	Interface et UX	HUD.tscn · Needle cooldown arc · LoreFragment modal · accessibilite mobile (44px min)
Game Designer 3	Level Design	Paper design 3 sous-zones · construction TileMap · courbe difficulte · arena boss
Full Stack	Level Design + Narration + Build	Beats narratifs · fragments lore · export Android/iOS/EXE · page itch.io · video gameplay
Dev 1	Developpement Godot	Player.gd · physique joueur · gravite inversee · coyote time · jump buffer · Needle
Dev 2	Developpement Godot	GameManager · Checkpoint · SceneTransition · AudioManager · ExitRift · LorePickup
Dev 3	Developpement Godot + Build	UnstableZone · integration Boss (BossAI.gd) · export pipeline · bug fix
Expert IA	Intelligence Artificielle	State machine Boss (IDLE/CHARGE/PULSE/SPLIT/DEATH) · balance · edge cases — aide par Dev 1, 2, 3

Game Analysis

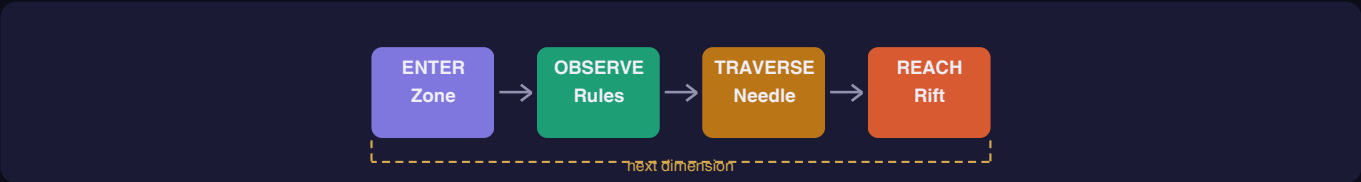


◆ STORY PREMISE

Long ago, all dimensions existed in separation. Then someone — or something — pulled on the thread. The fabric between worlds unraveled. Fragments of dimensions began to touch, merge, bleed into each other.

The Weaver belongs to the space between dimensions. Armed with the Needle, they travel through chaos — not to conquer, but to understand. Who caused the Great Tear? What should be done about it?

◆ CORE GAMEPLAY LOOP



- **Enter Zone** — Arrive in a new dimension via a tear. Physics rules immediately visible.
- **Observe Rules** — No tutorial text. Learn through exploration and failure.
- **Traverse with Needle** — Use the fold mechanic to navigate obstacles and puzzles.
- **Reach the Rift** — Cross the exit tear. Unlock next dimension.

Ref: Hollow Knight (env. learning) · Celeste (die-learn-retry) · Journey (transition as event)

◆ THEME — PLAY WITHOUT BORDERS

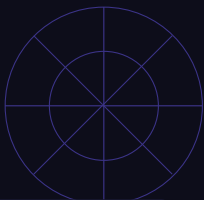
Interpretation	In-Game Implementation
Literal: crossing dimension borders	Each zone = crossing a physical border with new physics rules
Mechanical: borders as game system	The Needle manipulates dimensional seams — borders ARE the mechanic
Cultural: no cultural borders	African-inspired aesthetics (kente, bogolan, Sahelian) in a sci-fi universe
Narrative: repair or tear borders	Final choice: mend the borders or destroy them entirely



◆ DESIGN QUESTIONS

Question	Answer
Artistic genre	Oniric · Sci-fi · African-inspired fantasy · Dark contemplative
Win condition emotion	Awe and relief crossing the Rift — or dread if you tear it further
Emotions targeted	Wonder · Tension · Curiosity · Agency
Main loop duration	8–12 min / zone (jam) · 90 min total (full game)
The fun element	Mastering impossible physics · Precision Needle · Environmental storytelling
Game scenes	Intro → Main Menu → Zone (3 sub-zones) → Final Rift → Credits

Core Mechanics



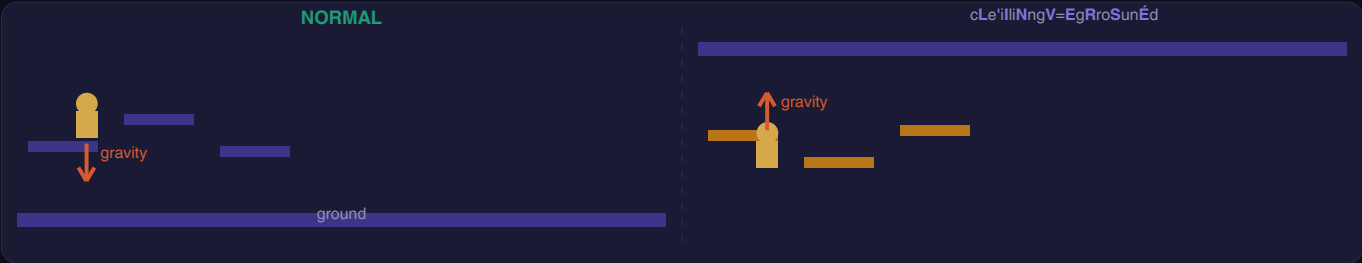
1 Inverted Gravity — L'Inversé

◆ DESCRIPTION

In L'Inversé, gravity pulls upward. The player falls toward the ceiling. Jumping means diving downward. Platforms hang from the sky. No tutorial — the environment teaches by doing.

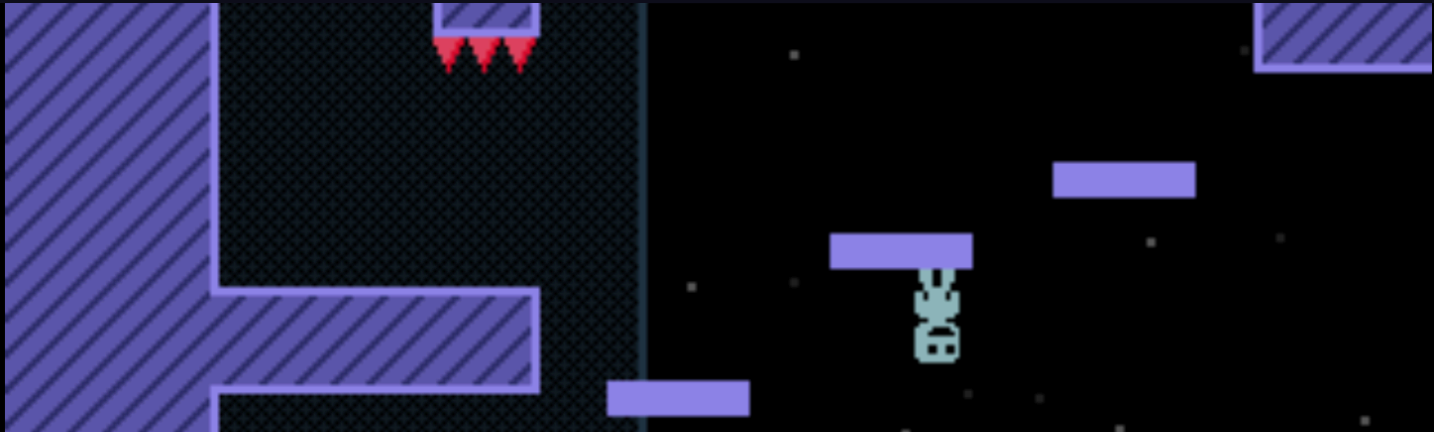
◆ RULES

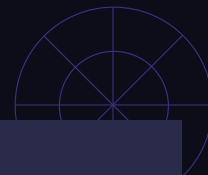
- Gravity reversed globally (**Godot: gravity_scale = -1**)
- Player **falls upward** — ceiling is the new ground
- Jump input sends player **downward** (context-inverted)
- All physics objects obey inverted gravity
- Unstable zones can appear on the floor (now above the player)



Left: normal world. Right: L'Inversé — player and objects fall toward the top of the screen.

◆ REFERENCE





Parameter	Value	Godot Implementation
Gravity scale	-1.0	CharacterBody2D.gravity_scale = -1.0
Jump direction	Downward	Invert jump vector Y axis
Coyote time	6 frames	Timer node — delayed on_floor check
Jump buffer	6 frames	Input buffer array, checked on landing

Ref: VVVVVV (gravity flip) · Super Mario Galaxy (local gravity)

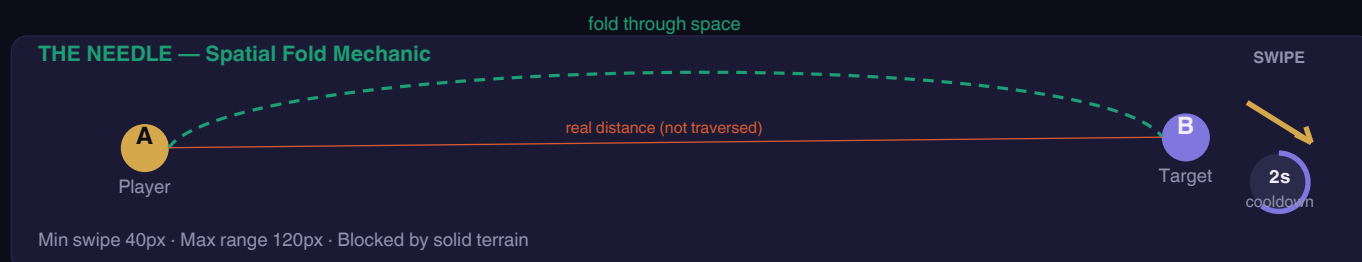
2 The Needle — Spatial Fold

◆ DESCRIPTION

The Weaver's primary tool. A swipe gesture creates a short-range fold in the dimensional fabric — the player is instantly displaced to a target point within range. Precise, tactical repositioning that also serves as the puzzle-solving mechanic.

◆ RULES

- Activated by **swipe gesture** — minimum threshold 40px
- Displacement range: **max 120px** from player position
- Cooldown: **2 seconds** — arc draining around the Needle icon
- Cannot fold into solid terrain — blocked fold shows rejection effect
- Can be used **mid-jump** — enables aerial repositioning
- Escaping an Unstable Zone edge with Needle = core skill expression



A → B: spatial fold (dashed arc). Red line = real distance — never traversed.



◆ REFERENCE



Parameter	Value	Godot Implementation
Input	Swipe (InputEventScreenDrag)	relative.length() > 40px threshold
Max range	120px	Distance check before teleport
Cooldown	2.0 seconds	Timer node + UI arc animation
Blocked feedback	Shake + glow pulse	AnimationPlayer on Needle icon
Mid-air use	Allowed	No floor check required

Ref: Hollow Knight (Dash with cloak of shadows)

Core Mechanics (continued)



3

Unstable Zones — Dynamic Threat

DESCRIPTION

Zones of decomposing dimensional fabric — visually red and vibrating. Contact = instant death. They expand slowly, creating urgency without a visible timer. The AI expert manages expansion speed, terrain pathfinding, and adaptive difficulty.

RULES

- Visual: pulsing red/crimson patches with particle edge distortion
- Contact = **instant death** → respawn at last checkpoint
- Default expansion: **2px per second**
- Expansion **blocked by platforms** — flows around terrain
- Player dies **3× at same checkpoint** → expansion drops to 1px/sec (AI adaptive)
- Two types: **static** (environmental hazard) and **dynamic** (active threat)



Zone expansion over time. AI slows rate after repeated deaths at same checkpoint.

REFERENCE



Parameter	Value	Godot Implementation
Shape	Polygon2D + Area2D	CollisionPolygon2D for contact
Expansion	2px/sec default	Tween on polygon vertices
AI adaptation	After 3 deaths: rate ×0.5	death_counter var per checkpoint
Terrain blocking	Yes	Raycast before vertex expansion

Ref: Hollow Knight (acid zones) · Celeste (auto-scroll) · Into the Breach (telegraphing)

4

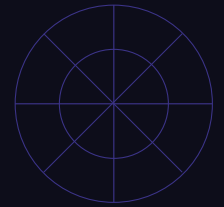
Checkpoint System — Couture Seams

◆ DESCRIPTION

Checkpoints are visual narrative elements — glowing couture stitches in the dimensional fabric. Touching one saves progress, plays a soft audio cue, and briefly stabilizes nearby Unstable Zones.

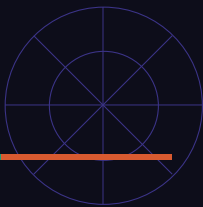
◆ RULES

- Triggered by **Area2D overlap** — no button press needed
- Saves: player position, zone state, collected lore fragments
- Visual: stitching glow animation (1 second on activation)
- Nearby Unstable Zones **pause expansion 3 seconds** on activation
- **3 checkpoints** total in the prototype zone



Checkpoint	Location	Sub-zone unlocked	Special
CP-1	End of sub-zone 1	Sub-zone 2	Tutorial complete moment
CP-2	Mid sub-zone 2	—	Lore fragment triggers here
CP-3	Start of sub-zone 3	Final Rift approach	Zone visually destabilizes

Core Mechanics (continued)

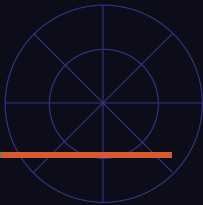


◆ MOBILE CONTROLS — PORTRAIT LAYOUT

Input			
Action		Technical Note	Priority
Move left/right	Virtual joystick — left thumb	Dead zone 0.2	Must
Jump / Dive	Button A — right thumb	Y-inverted in L'Inversé	Must
Needle fold	Swipe in direction	40px min threshold	Must
Inspect fragment	Tap on collectible	Area2D tap detection	Nice
Pause	Tap top-right icon	Dark overlay, instant	Must

Critical mobile feel: implement **coyote time (6 frames)** and **jump buffer (6 frames)** in Godot — without these, the game will feel unjust on touchscreen and lose points on 'intuitive gameplay'.

World & Dimensions



◆ PROTOTYPE ZONE — L'INVERSÉ

Element	Description
Visual palette	Deep violet, gold, luminous white — kente & indigo inspired
Music	Reversed ambient samples, distant bass drums, no melody
Zone structure	3 sub-zones separated by checkpoint seams (glowing stitches)
Playtime	8–12 min first run · 4–5 min speedrun
Climax	The Great Rift — pulsing portal that widens. No boss combat.
Win condition	Reach the Rift and cross it (+ optional choice: repair/tear)
Fail condition	Touch Unstable Zone · Fall out of bounds

◆ ART DIRECTION — AFRICAN-INSPIRED AESTHETIC

Element	Reference & Implementation
Platforms & terrain	Kente cloth geometric patterns (Ghana) — interlocking diamonds and chevrons in gold/purple
Color palette	Bogolan/mud cloth (Mali) — deep indigo, earthy terracotta, warm ochre
Background arch.	Sahelian mosque silhouettes (Djenné, Agadez) — organic clay spires, banco walls
The Weaver	Tall slender silhouette, woven garment shifting color per dimension
The Needle	Elongated ivory/bone with geometric carvings — glows teal at use
UI / HUD	Minimal — Needle cooldown arc, health as thread unraveling, no numbers

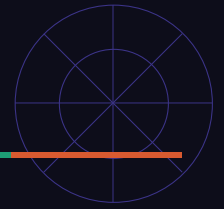
◆ ALL SIX DIMENSIONS — FULL VISION

Dimension	Core Rule	Jam Status
L'Inversé	Gravity pulls up. Objects fall upward.	Prototype — Must
Le Muet	No sound. Thoughts appear as geometric shapes.	Bonus if time allows



Le Liquide	Floors flow. Faster you move, slower you go.	Post-jam
Le Brûlant	Colors have heat. Red burns, blue freezes.	Post-jam
L'Ancien	Time flows backward. Ruins rebuild.	Post-jam
Le Miroir	Everything mirrored. Built for multiplayer.	Post-jam (multi)

Story & Narrative



◆ STORY TREATMENT — PROTOTYPE (L'INVERSÉ)

Intro (pre-menu)

Static shot of a pulsing tear in reality. Silence. A faint sound — something being pulled. The Weaver falls upward through the screen. Cut to title.

Sub-zone 1 (learning)

Player discovers the inversion. No text. A platform that should be below is above. First Unstable Zone in the floor. CP-1 reached — soft sound, visual calm.

Sub-zone 2 (lore)

A floating shard near CP-2. The Weaver touches it. Brief vision: who lived here before the inversion. 3 seconds of imagery. No words.

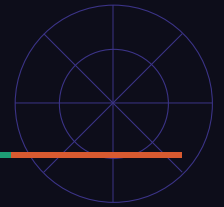
Sub-zone 3 (climax)

The Great Rift pulses and widens. Unstable Zones expand faster. The zone visually collapses. The Weaver reaches the Rift.

Final choice

Two gestures on screen — Needle inward (repair) or outward (tear further). Two visual endings. No text. Player interprets.

11-Day Development Plan



◆ SPRINT TIMELINE

Days 1–2

Setup & Movement Prototype

- Godot repo + scene structure (Main, Zone_Inversé, UI, GameManager)
- CharacterBody2D: movement + inverted gravity (gravity_scale = -1)
- Placeholder tileset — geometry only, no art yet
- Basic death zone (red Area2D trigger)
- ✓ Deliverable: player falls UP, no crash, no clip through ceiling

Days 3–4

Needle + Unstable Zones

- Swipe input: InputEventScreenDrag + 40px threshold + fold teleport
- Cooldown 2s + UI arc animation on Needle icon
- Unstable zone Polygon2D expansion via Tween (AI system)
- Death + respawn at last checkpoint
- Coyote time (6f) + jump buffer (6f)
- ✓ Deliverable: full die/respawn/needle loop working

Days 5–6

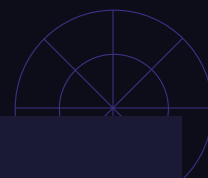
Level Design + Art Pass 1

- Paper design first: 3 sub-zones, platforms, zones, checkpoints
- Build from paper design in Godot
- Cartoon tileset: kente/indigo palette — platforms, background, terrain
- Character animations: idle, run, jump, needle-use, death
- Checkpoint glow animation (stitching effect)
- ✓ Deliverable: zone playable end-to-end

Days 7–8

Lore + Sound + Juice

- 3–5 lore fragment images (static illustrated visuals)
- Ambient music: reversed samples + bass drums
- SFX: Needle activation, checkpoint, death, zone hum
- Particle FX: zone edges, needle fold flash, checkpoint stitch
- Title screen + end screen + final choice (if time allows)
- ✓ Deliverable: complete experience from title to credits



Days 9–10

Playtests + Polish

- 5+ external playtests — NOT team members
- Log: death locations + Needle comprehension
- Tune: expansion speed, Needle range, checkpoint placement
- Critical bug fixes only — NO new features after day 8
- Mobile build tested on real device (iOS + Android)
- ✓ Record 60–90s gameplay video + portrait screenshots

Day 11

Submission

- EXE build + APK/IPA finalized and tested
- itch.io page: description + African aesthetic + double theme
- GDD submitted with all required assets
- ■ 4-hour buffer — NO code changes on submission day

Scope rule: Day 6 and zone not playable → cut lore + final choice. Day 8 and no sound → ship silent.
Non-negotiable: inverted gravity + Needle + 1 complete zone on mobile.

Mechanic Details — What Exactly Gets Built



This section is for the team. Each mechanic is explained in plain terms — what the player sees, the exact values, and what a correct implementation looks like. If you have a doubt about what is expected from you: the answer is here.

◆ THE PROTOTYPE IN ONE SENTENCE

The player arrives in L Inverse (gravity pulls up), learns the rules, uses the Needle to cross obstacles, avoids expanding Unstable Zones, hits 3 checkpoints, and reaches the Exit Rift. That is the complete loop. Nothing else is required for the jam.

◆ WHAT THE PLAYER MUST BE ABLE TO DO

- Move left/right with the virtual joystick
- Jump (downward in inverted mode) with button A
- Use the Needle by swiping in a direction on screen
- Die when touching an Unstable Zone and respawn at the last checkpoint
- Activate a checkpoint by touching it — no button press
- Reach the Exit Rift and see an end screen

◆ OUT OF SCOPE FOR THE JAM

- Other dimensions (Le Muet, Le Liquide...) — post-jam only
- Combat / enemies — post-jam only
- Multiplayer — post-jam only



◆ MECHANIC 1 — INVERTED GRAVITY: HOW IT WORKS IN PLAIN TERMS

In L Inverse, gravity pulls the player upward. The ceiling is the floor. Platforms hang below in the normal sense, but the player falls toward them from above. Pressing JUMP propels the player downward. The zone teaches by doing — no tutorial text.

Mental image: the level is flipped 180 degrees. What was up is now down. The player falls toward what was the ceiling.

◆ INVERTED GRAVITY — EXACT VALUES

Parameter	Value	Why it matters
gravity_scale	-1.0	Set on CharacterBody2D — reverses Godot gravity direction
Default gravity	980 px/s	Godot default — do not change
Move speed	200 px/s	Horizontal only — same in normal and inverted mode
Jump force	400 px/s	Applied downward in inverted mode: $\text{velocity.y} = +400.0$
Max fall speed	600 px/s	Clamp velocity.y — prevents infinite acceleration
Coyote time	6 frames 0.1s	Player can jump 6 frames after leaving a platform edge
Jump buffer	6 frames 0.1s	Jump input remembered 6 frames before landing

Why coyote time and jump buffer are non-negotiable: on mobile, inputs are imprecise. Without these two systems the player feels the game ignores their inputs. This is the difference between frustrating and satisfying. Standard in Celeste, Hollow Knight, Mario.

◆ MECHANIC 2 — THE NEEDLE: HOW IT WORKS IN PLAIN TERMS

The player swipes in a direction. The Weaver teleports instantly up to 120px that way. Not a long portal — a short tactical displacement. If terrain blocks the path, rejection effect plays.

Mental image: swipe right and appear 120px to the right instantly. Like a teleport dash. Swipe direction = fold direction.

◆ NEEDLE — STEP BY STEP

- Player swipes on screen (minimum 40px to trigger — avoids accidents)
- Game calculates target: player position + direction x 120px
- Raycast toward target on terrain layer (layer 1 only)
- If clear: teleport, reset velocity to zero, start 2s cooldown
- If blocked: rejection animation on Needle HUD icon + rejection sound
- During cooldown: Needle icon greyed out, cannot use

◆ NEEDLE — EXACT VALUES

Parameter	Value	Detail
Max range	120 pixels	If wall before 120px, teleport stops at wall
Min swipe	40 pixels	Below 40px: no trigger, prevents accidents
Cooldown	2.0 seconds	Timer before Needle can be used again
Velocity after fold	Vector2.ZERO	Player speed reset to zero after teleport
Raycast layer	Layer 1 terrain	Only solid terrain blocks — not Unstable Zones
Mid-air use	Allowed	Player does not need to be on the ground

◆ NEEDLE — ROLE IN THE LEVEL

- Sub-zone 1 Discovery: a gap too wide to jump forces the first swipe. No explanation, just a gap.
- Sub-zone 2 Mastery: puzzles where the Needle is the only way past an Unstable Zone.
- Sub-zone 3 Urgency: fast reaction use to escape rapidly expanding zones.



◆ MECHANIC 3 — UNSTABLE ZONES: HOW THEY WORK IN PLAIN TERMS

Red vibrating patches of decomposing fabric. Contact = instant death. They expand slowly, creating urgency without a visible timer. The AI system manages growth and adapts to the player deaths.

Mental image: lava in Mario, but it grows. Longer the player takes, more it spreads. Urgency comes from time, not from attacking enemies.

◆ UNSTABLE ZONES — EXACT VALUES

Parameter	Value	Detail
Expansion rate normal	2.0 px/s	Grows outward from center in all directions
Expansion rate adapted	0.8 px/s	After 3 player deaths at same checkpoint
Death threshold	3 deaths	Deaths before AI reduces expansion
Checkpoint pause	3.0 seconds	All zones within 300px pause when checkpoint activates
Terrain blocking	Yes	Zones flow around walls, never pass through terrain
is_dynamic = true	Dynamic zone	Grows over time — main threat type
is_dynamic = false	Static zone	Fixed size — environmental hazard, no growth

◆ PLACEMENT IN L INVERSE

Sub-zone	Count	Type	Purpose
Sub-zone 1	1	Static	Introduce concept without time pressure
Sub-zone 2	2	Dynamic	Create urgency, force Needle use
Sub-zone 3	3 (1 fast)	Dynamic — 1 at 3.0 px/s	Maximum tension at climax

◆ MECHANIC 4 — CHECKPOINTS: HOW THEY WORK IN PLAIN TERMS

Glowing stitches in the dimensional fabric. Touching one saves position. Die after: respawn there, not at start. One touch activates permanently. No button press.



◆ CHECKPOINTS — WHAT HAPPENS ON ACTIVATION

- Player enters Area2D: body_entered fires (no button needed)
- Glowing stitch animation plays — 1 second
- Soft audio cue: thread pulling sound
- GameManager.player_spawn_position = player current position
- All Unstable Zones within 300px pause expansion for 3 seconds
- Checkpoint cannot be deactivated — one-time activation only

Checkpoint	Position	What changes after
CP-1	End of sub-zone 1	Player masters inverted gravity and just found the Needle
CP-2	Mid sub-zone 2	Calm moment — lore fragment appears here
CP-3	Entry sub-zone 3	Zone becomes visually unstable — tension builds to climax

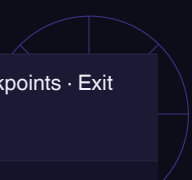
◆ PROTOTYPE DEFINITION — WHEN IS IT DONE?

A jam prototype does not need to be perfect. It needs to be playable from start to finish. One zone, one complete loop, no crashes.

Done when an external player — who has never seen the game — can:

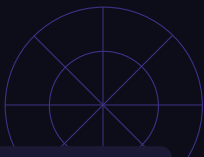
- Launch on Android or iOS
- Understand movement and Needle without anyone explaining
- Cross all 3 sub-zones of L Inverse
- Die, respawn at the checkpoint, and continue
- Reach the Exit Rift and see an end screen

Category	Must have
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Gameplay	Inverted gravity + coyote time + jump buffer · Needle swipe + 2s cooldown · 3 zones (2 dynamic) · 3 checkpoints · Exit Rift
UI	Virtual joystick left · Jump button right · Needle cooldown indicator · End screen
Audio	Background music (royalty-free OK) · Death SFX · Checkpoint SFX
Art	Player: idle + run states · Distinct platform tiles · Readable Unstable Zone (red)

Role Deliverables — Who Does What



Exact list of what each member must deliver. No ambiguity. If a task is not on your list, it is not your jam priority.

◆ GAME DESIGNERS 1 & 2 — INTERFACE & PLAYER EXPERIENCE

Responsible for everything the player interacts with that is NOT the world itself: menus, HUD, feedback visuals, and the overall feel of the UI.

Day	GD 1	GD 2
Day 1	Wireframe MainMenu (paper or Figma): logo, PLAY button, SETTINGS button, version label	Wireframe HUD (paper or Figma): joystick zone, jump button, Needle cooldown arc, health thread
Day 2	MainMenu.tscn in Godot: layout, buttons connected to SceneTransition	HUD.tscn in Godot: virtual joystick left, button A right, TextureProgressBar Needle arc
Day 3	PauseMenu.tscn: RESUME + QUIT buttons, dark overlay, pause toggle on tap top-right	HUD.gd: connect Needle cooldown_started signal → tween arc value 100→0 over 2s
Day 4	Settings screen: volume slider, language toggle FR/EN	HUD.gd: connect checkpoint_reached signal → brief flash/pulse on screen edge
Day 5	EndScreen.tscn: final choice UI (Repair / Tear) — two gesture prompts or two buttons	LoreFragment.tscn: modal overlay — image + close tap anywhere
Day 7	Polish pass: button animations, transitions feel, font consistency across all screens	Accessibility: ensure all tap targets are minimum 44x44px for mobile fingers
Day 9	Final UI review — test all screens on real device, no overflow or clipping	Prepare 5 portrait screenshots 390x844 showing UI in action for itch.io

Key rule for UI: every tap target minimum 44x44px. Test on a real phone before day 9 — what looks fine in the Godot editor often breaks on device.

◆ GAME DESIGNER 3 & FULL STACK — LEVEL DESIGN + NARRATIVE

GD3 and Full Stack share the level. GD3 owns the spatial design (platform placement, zone layout, pacing). Full Stack owns the narrative layer (story beats, lore fragments, tone of the zone) and handles the build pipeline.

Day	GD 3 — Level Design	Full Stack — Narrative + Build
Day 1	Paper design of all 3 sub-zones: platform positions, gap sizes, checkpoint locations, zone instable spots	Git repo setup: branches main/dev/feature-xxx, add all team, write README
Day 2	Validate paper design with Dev 1 — check gaps are jumpable, Needle puzzles are solvable	Write narrative beats: intro (pre-menu), sub-zone 1 discovery, CP-2 lore vision, climax, final choice
Day 3	Translate paper design to Godot TileMap — sub-zone 1 complete with placeholder tiles	Write lore fragment content: 3-5 images described in detail for the artist to illustrate
Day 4	Sub-zone 2 in TileMap — place 2 dynamic Unstable Zones, design Needle puzzle	Android export preset configured (API 21+, arm64) — APK installable on device
Day 5	Sub-zone 3 in TileMap — place 3 Unstable Zones + boss arena at climax before Exit Rift	Verify project runs on all team OS (Windows/Mac/Linux)
Day 6	Full zone walkthrough test: is it fair? Is the difficulty curve correct? Adjust platform gaps.	itch.io page draft: description with Play Without Borders double meaning, African team, GameCloud
Day 9	Final level polish: visual readability, no softlocks, no sequence breaks	Record 60-90s gameplay video for submission — screen record, no voice needed

Level design rule: design on paper **FIRST**, then build in Godot. A gap that seems fine on paper might be impossible with inverted gravity. Validate every puzzle with Dev 1 before building it in TileMap.

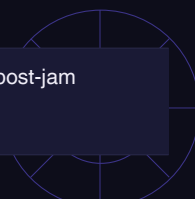
◆ DEV 1 — PLAYER PHYSICS & CORE MECHANICS

Responsible for everything the player character does: movement, gravity, the Needle, death, respawn.

Day	Task	Deliverable
Day 1	Player.tscn: CharacterBody2D + CapsuleShape2D + AnimatedSprite2D + AnimationPlayer. Player.gd: horizontal movement 200 px/s, basic gravity, test scene.	Player moves left/right and falls down
Day 2	Inverted gravity: gravity_scale = -1.0 applied by Zonelnverse. is_on_ceiling() as floor detection. set_gravity_inverted(bool) function.	Player falls UP in L Inverse
Day 2	Coyote time: 6 frames counter in _physics_process (not a Timer node). Jump buffer: 6 frames input memory.	Jumps feel responsive on mobile
Day 3	PlayerNeedle.gd: InputEventScreenDrag (40px min) + direction + raycast layer 1 + teleport + velocity reset + 2s cooldown Timer.	Needle works on mobile swipe and desktop right-click
Day 4	Player.die(): is_dead=true, animation, await 1s, respawn(). Player.respawn(): position = GameManager.player_spawn_position, velocity reset.	Die and respawn loop functional

Day 5 ZoneBase.gd: parent class with apply_zone_rules(player) — override per dimension. ZoneInverse.gd: calls set_gravity_inverted(true).

Architecture ready for post-jam dimensions



End of Day 2 target: player falls UP, coyote time and jump buffer working, no clipping through ceiling. If this is not done by end of Day 2, all other systems are blocked — this is the critical path.

◆ DEV 2 — SYSTEMS, CHECKPOINTS & TRANSITIONS

Responsible for all transversal systems: GameManager, checkpoints, scene transitions, Unstable Zone signals, and audio management.

Day	Task	Deliverable
Day 1	GameManager.gd (autoload): player_spawn_position, death_counter dict {cp_id: int}, signals player_died / checkpoint_reached / zone_entered / lore_collected.	GameManager ready for all systems to connect
Day 1	InputHandler.gd (autoload): swipe_detected signal, touch start/end tracking per finger index.	Unified input layer for mobile and desktop
Day 2	Checkpoint.gd: Area2D body_entered → GameManager.set_checkpoint(id, pos) → notify UnstableZones within 300px → pause_expansion(3.0).	Checkpoints save and pause zones
Day 3	SceneTransition.gd (autoload): fade_to_black() and fade_from_black() with AnimationPlayer. Transition between MainMenu and ZoneInverse.	Smooth transitions between scenes
Day 4	AudioManager.gd (autoload): play_music(stream), play_sfx(stream), stop_music(), set_volume(). Preload ambient track and SFX.	Music and SFX play correctly
Day 5	ExitRift.gd: Area2D body_entered → SceneTransition → EndScreen. Signal game_completed(choice) to GameManager.	Exit Rift ends the zone
Day 6	LorePickup.gd: Area2D body_entered → GameManager.collect_lore(id) → show LoreFragment modal via signal.	Lore collection works

◆ DEV 3 — AI INTEGRATION & BUILD PIPELINE

Responsible for integrating the AI Expert work into Godot, the Unstable Zone system, and all export/build tasks.

Day	Task	Deliverable
Day 1-2	UnstableZone.tscn: Area2D + CollisionPolygon2D + Polygon2D (octagon ~50px). Work with AI Expert on expansion algorithm structure.	Zone structure ready for AI Expert to implement behavior

Day 3	Integrate AI Expert expansion code: <code>_expand(delta)</code> , terrain raycast blocking, <code>body_entered</code> death detection.	Zones expand and kill on contact
Day 4	Boss scene scaffold: <code>Boss.tscn</code> placeholder, <code>BossAI.gd</code> stub with state machine structure. Work with AI Expert on state definitions.	Boss scene exists and can be populated by AI Expert
Day 5	Android export preset: API 21+, arm64, debug keystore. Test APK on real device.	APK installable and running on Android
Day 6	Integrate Boss full behavior from AI Expert. Place Boss in sub-zone 3 arena (from GD3 level design).	Boss active in level
Day 8	Windows EXE export + iOS export if possible. Verify all builds on team machines.	All builds functional
Day 9	Bug fix integration: collect team bug reports, triage and assign fixes.	Build stable for playtests

◆ EXPERT IA — BOSS & HOSTILE MOBS

Responsible for the intelligence of all hostile entities. Works closely with Dev 3 (integration) and the Devs for Godot-specific implementation questions. For the jam: one boss at the climax of L Inverse.

Jam scope for IA: 1 boss at the climax of sub-zone 3 (before Exit Rift). No roaming mobs for the prototype — too risky in 11 days. Mobs are post-jam.

◆ BOSS — THE RIFT GUARDIAN

A fragment of the dimensional fabric that became sentient when the Great Tear happened. It does not want the Weaver to leave. It manifests as a shifting mass of unstable energy that attacks by sending pulses of instability across the arena.

State	Trigger	Behavior	Duration
IDLE	Boss spawns	Hovers in center of arena, slow oscillation	2 seconds
CHARGE	Player enters arena	Boss moves toward player position at 80 px/s	Until near or 3s
PULSE	Within 100px of player OR every 4s	Emits 3 expanding instability rings outward — player must Needle dodge	1.5 seconds
SPLIT	Health below 50%	Splits into 2 smaller fragments with same state machine	Permanent
DEATH	Health = 0	Collapse animation, arena clears, Exit Rift becomes accessible	3 seconds

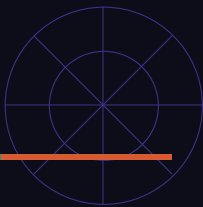
◆ BOSS — TECHNICAL SPECS



Parameter	Value	Note
Max health	6 hits	Each Needle hit = 1 damage. No melee — Needle only.
Move speed	80 px/s	In CHARGE state — respects inverted gravity
Pulse speed	150 px/s	Instability rings expand outward — contact = player die
Pulse cooldown	4.0 seconds	Between pulse attacks
Hitbox	CircleShape2D r=30px	Needle raycast must intersect — not Area2D
SPLIT threshold	50% health (3 hits)	Splits into 2 at speed 60 px/s each
AI update rate	_physics_process	Full logic every frame — no Timer-based AI

Day	Task	Works with
Day 1-2	Design boss state machine on paper: states, transitions, triggers. Write pseudocode for each state.	GD3 (arena design)
Day 3	Implement state machine in BossAI.gd: IDLE and CHARGE states. Basic pathfinding toward player.	Dev 3 (integration)
Day 4	PULSE state: emit 3 expanding CircleShape2D rings. Player contact = death. Ring lifetime 2s.	Dev 1 (player hitbox)
Day 5	SPLIT state: spawn 2 child Boss instances at half health, each with independent state machine.	Dev 3 (scene instancing)
Day 6	DEATH state: collapse animation → signal boss_defeated → Dev 2 enables Exit Rift.	Dev 2 (ExitRift signal)
Day 7	Balance pass: is the boss fair with inverted gravity? Adjust speeds, pulse frequency, health.	GD3 (playtesting in arena)
Day 8	Edge cases: what if player dies during SPLIT? What if player Needles out of arena bounds?	Dev 1 + Dev 3

Scoring Strategy



◆ SCORE BREAKDOWN & STRATEGY

Criterion	Pts	Our Advantage	Risk
Game Prototype	60	Clear loop · original mechanics · native mobile controls + coyote time	Mobile calibration in 11 days
Theme Adoption	20	Double reading: literal (dimension borders) + metaphorical. 'Borders' = seams = core mechanic.	None — strongest point
Creativity	10	African aesthetic in sci-fi = globally rare. Env. lore with no text.	Ambitious art in 11 days
Commercial	10	Post-jam: 6 dimensions, multiplayer, seasonal arcs. Target: mobile 16–30.	Competitive mobile market
Bonus Cultural	+5	African team + kente/bogolan/Sahelian aesthetics integrated into gameplay.	Must be explicit in submission

◆ REFERENCES

Game	Developer	What We Take
Hollow Knight	Team Cherry	Environmental storytelling · atmospheric platformer · lore without dialogue
Celeste	Maddy Thorson	Precise mobile controls · coyote time · difficulty through environment
Manifold Garden	William Chyr	Impossible physics per zone · architecture as puzzle
Gris	Nomada Studio	Wordless narrative · color as world state · contemplative platformer
Journey	thatgamecompany	Transition as aesthetic event · silence as language
VVVVVV	Terry Cavanagh	Pure gravity inversion — direct reference for L'Inversé